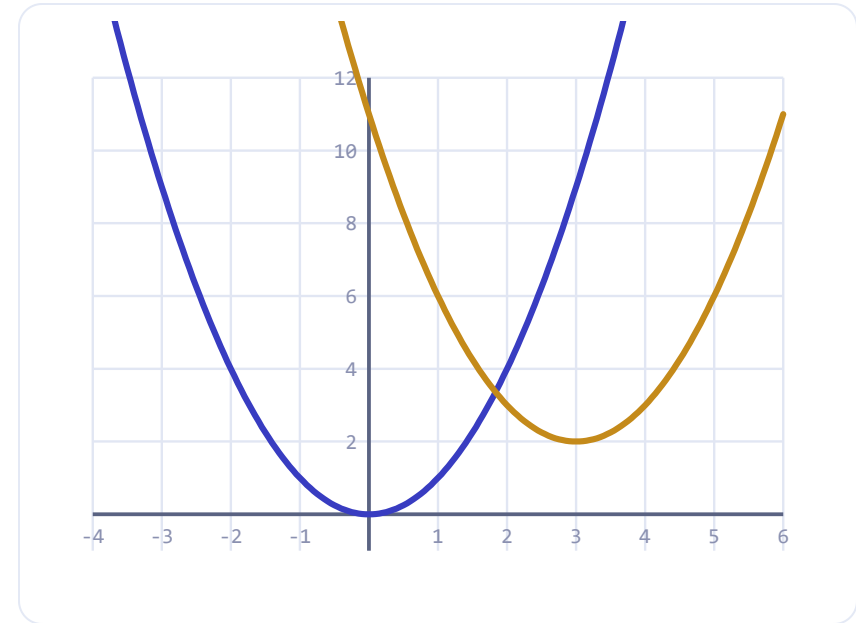


Same U, new address

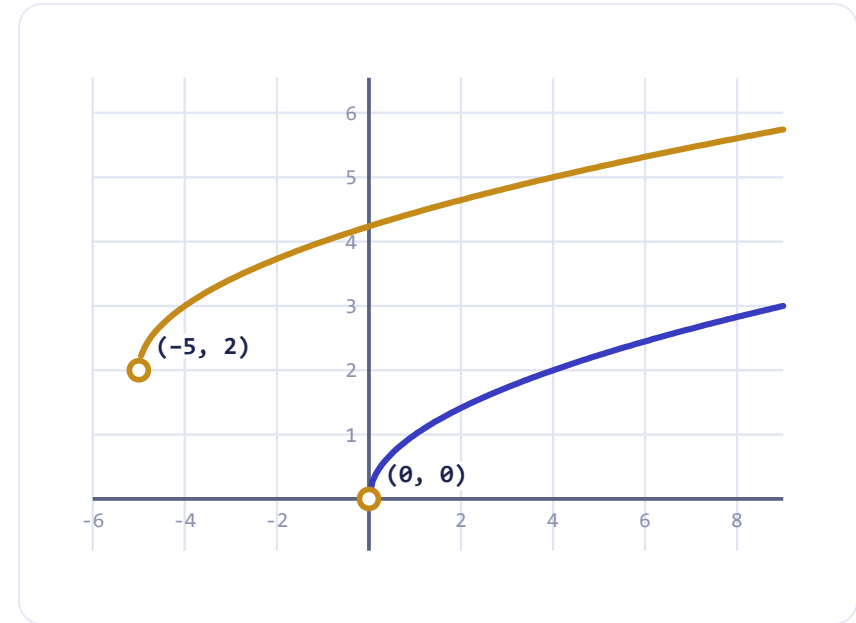
- ◆ Both of these are $y = x^2$ — **identical shape**.
- ◆ One of them just... moved. Where did it go, and what in the equation moved it?
- ◆ By the end of class you'll slide ANY parent anywhere on the grid.



$y = x^2$ (indigo) and its twin (gold) — same shape, new address

The two shift rules

- ◆ **Outside** the function → vertical, **as written**: $f(x) + k$ up, $f(x) - k$ down.
- ◆ **Inside** the function → horizontal, **opposite**: $f(x - h)$ RIGHT, $f(x + h)$ LEFT.
- ◆ The shape never changes — only the address.



$y = \sqrt{x}$ slid left 5, up 2 — the start point carries the whole graph

Today's words

Transformation — any change to a parent's graph

Translation — a slide — shape and size unchanged

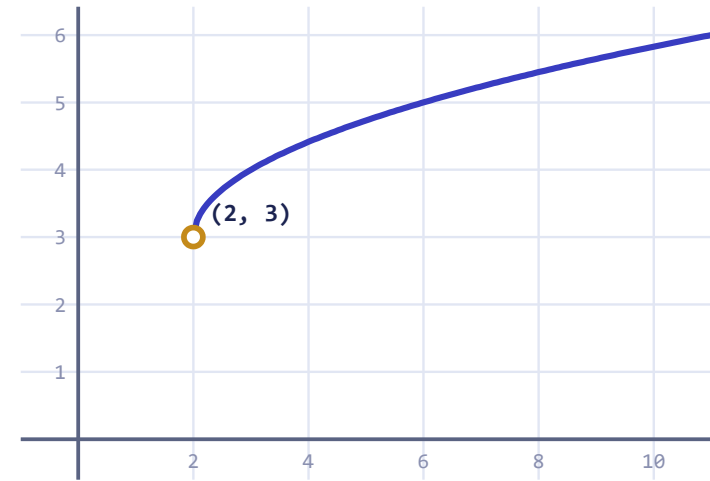
WORKED EXAMPLE

We do: $y = \sqrt{(x - 2)} + 3$

4 / 7

WORKED EXAMPLE

1. Inside: $x - 2 \rightarrow$ RIGHT 2 (inside is opposite).
2. Outside: $+3 \rightarrow$ UP 3 (outside is as written).
3. Start point: $(0, 0) \rightarrow (2, 3)$.
4. Domain $x \geq 2$; range $y \geq 3$ — they slide with the graph.



$$y = \sqrt{(x-2)} + 3$$

WATCH OUT

5 / 7

The inside sign lies

COMMON MISTAKE

$y = (x - 3)^2$ moves **RIGHT** 3 — not left.

The inside asks "what x makes this zero?" $x = 3$ does, so the vertex now lives at $x = 3$.

KEEP IN MIND

Read the inside as: the value that makes it zero is the new home.

YOU TRY

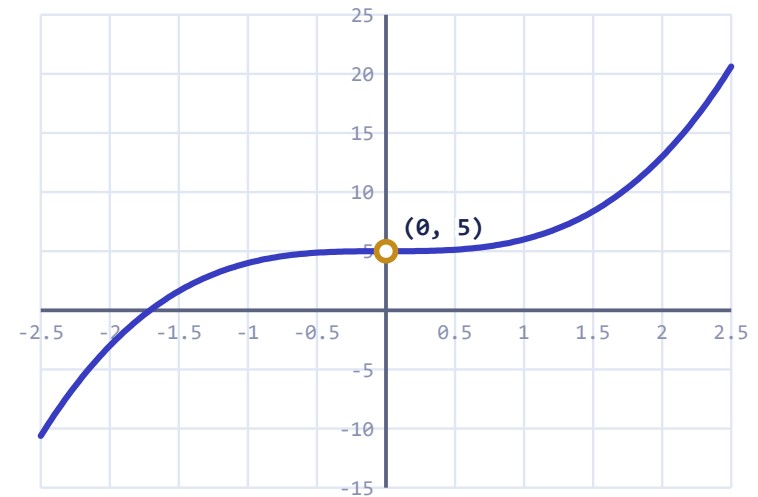
Your turn

6 / 7

YOU TRY

(a) Describe every shift: $y = x^3 + 5$

(b) Write the equation: $y = |x|$ shifted right 3



(a) to check against

Where this week goes

- ◆ **Tue** — reflections and stretches (the a out front).
- ◆ **Wed** — combine moves + write equations from a graph.
- ◆ **Thu** — graph any transformed parent from scratch.
- ◆ **Fri** — mixed practice + quiz-style check.